1.5 Living with robots

In this lesson

You will learn:

- how robots are changing the world you live in.

You have learned how robots help to make things in manufacturing industries. Robots and control systems make life easier in homes. Robots are used in many other industries.

Robots on farms

Harvesters, ploughs and other farm machines are driven by robots. The vehicles find their way around fields using the same **satellite navigation** (satnav) that car drivers use. The farmer does not have to drive farm vehicles so has time to do other tasks on the farm.

A **drone** is a helicopter or plane that does not need a pilot. Drones are robot-controlled. Drones are used to spray pesticides or fertilisers where they are needed.



Robot arms are used to harvest and pack produce. Robots use sensors to decide what is ripe. Robot arms with pressure sensors can pick delicate fruits and vegetables without causing damage.

Shopping with robots

More people buy products online. Online shops store the products in large warehouses. Robots find and pack items when they are purchased.

Online stores are experimenting with using robot vehicles and drones to deliver goods. Some high street stores are using robot shop assistants. They move around the shop and answer any questions that customers have.



Robots and jobs

Some people worry that robots will replace jobs and cause unemployment.

Other people say that robots and technology will replace some jobs but will also create new ones.

- Robots are good at repetitive tasks that need to be done quickly and accurately. Many of the jobs that robots replace are low-skilled. People often find the jobs stressful and boring.
- People are needed to invent and build new robots. New factories are needed for robot-controlled machines. **Computer technicians** are needed to install robots. **Software engineers** write programs to control robots. Technicians are needed to maintain and repair robots. The new jobs are often high-skilled and interesting.

Some people say introducing robots will create more new jobs than are lost.



Activity

Have a look at the two car assembly lines. One is from the 1930s when humans did all the work. The other is a modern assembly line with robots. Describe the differences you can see between the two pictures.





Think again

Which of the two assembly lines would you prefer to work in? Explain why.



Extra challenge

Do research on robots working in agriculture or retail. What kind of robot vehicles and machines are used in these industries?

1.6 Robots of the future

In this lesson

You will learn:

- how robots will develop in the future
- how robots are helping doctors
- how robots will change the way we travel



Artificial intelligence

Robots are being developed that can act more like humans. This is called **artificial intelligence** (AI). AI allows robots to:

- recognise pictures and speech
- learn better ways of doing things
- decide the best way to deal with situations.

In the future, a robot shop assistant may recognise how a customer feels from their voice and face. The robot will take a different approach depending on how the customer looks and sounds.

In the classroom, a robot assistant will learn about your progress and how you study. The assistant will set you individual projects and find learning materials that are best for you.

Robot vehicles



Robot cars are also called autonomous cars. Autonomous means the car works automatically with no need for any human help. An autonomous car does not need a driver. It has sensors that can detect pedestrians, other cars and road signs. The controller uses the information from the sensors to operate the accelerator, brakes and steering.

At any moment there will be different ways a robot car can deal with a situation on the road. For example, does it brake or swerve to avoid a collision? AI will help the car make the best decision.

Cars in the future will look different to cars today. An autonomous car does not need a steering wheel or pedals. Seats do not all need to face forward.

Robots in healthcare



Robots are already being used in modern hospitals. Scanners create a 3D picture of the inside of our bodies so that doctors can diagnose problems. Robots help doctors carry out delicate surgery they could not do without robotic instruments. Robots help doctors but will not replace them.

Robot nursing assistants carry out basic health checks on patients. These robots already help look after elderly patients who need regular care. A robot called Robear can lift patients in and out of their bed or wheelchair. It is more comfortable for the patient and prevents nurses from being injured.

Scientists are researching **nanobots**. A nanobot is a microscopic robot. In the future, nanobots may be able to work inside a patient's bloodstream to fight infections or carry out delicate surgery.



Activity

Travelling in cars is one of the most dangerous things we do. Will autonomous cars make car travel safer or more dangerous? Discuss this in a small group, then write down your own opinions.



Explore more

Choose one use for robots you have learned about in this unit. For example, robots in the home or robot cars. For the area you have chosen, find:

- examples of how robots are used
- photographs of robots
- interesting facts and figures.

Go back over the content of this unit and do extra research using the web.

Present your findings as a presentation, word-processed document or website.



Extra challenge

Carry out more research on either robot cars or robots in healthcare. Make notes. Find examples, facts and pictures.

